



Design of an HMI System Promoting Smooth and Safe Transition to Manual from Level 3 Automated Driving

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Abstract

We designed an HMI system to induce a smooth and safe transition to manual driving from Level 3 automated driving with using voice guidance, alarm sound, HUD and interior lights. On the route to the transition zone, the HMI system supplies two preparation zones to regain the awareness of driving and speed. An experiment and an investigation clarified that the HMI system is more effective to induce smooth and safe transition to manual driving than another HMI system for comparison with only alarm sound and HUD. It is also suggested that preparation zones are important for drivers to take over driving readily with concentration.

Keywords Level 3 automated driving system · Transition of driving · Human machine Interface

1 Introduction

Automated driving systems are divided into five levels according to automation levels proposed by the Society of Automotive Engineers (SAE). In Level 1 and 2 vehicles, drivers must be engaged with the driving task and monitor the environment at all time. On the other hand, they are not required to monitor in Level 3 through 5 vehicles. The drivers can do what they want like eating or reading while automated driving systems taking on driving functions. However, in a level 3 vehicle, the driver must take over controlling the vehicle when the system requests [1]. Although the driver is expected to be always receptive to the request, it is not easy for him to stay alert to the request. It is also difficult for him to take back drive sense including sense of speed instantly. Thus, a transition to manual driving from Level 3 automated driving is an important issue to be addressed [2–6].

Human Machine Interface (HMI) systems are effective to induce drivers' safe driving [7–12]. It is also expected that HMI systems will promote smooth and safe transition to manual from level 3 automated driving.

2 Objective

In the present study, we designed an HMI system to guide drivers so that they can take back driving smoothly and safely from Level 3 automated driving. We, then conducted an experiment and an investigation to examine the effect of the HMI system.

3 Methodology

There is a study based on the similar purpose by Homma et al. of the Japan Automobile Research Institute (JARI) [13]. They studied on transition from automated to manual driving in the situation that the system detected a lane closer. A warning sound and head up display (HUD) were used to prompt drivers to take over driving. The drivers were prompted 2, 5 and 10 s before the vehicle reached the road closer. Through the experiment, the following conclusions were reported. It is difficult for drivers to take over driving safely in 2 s. Even if drivers have 5 or 10 s to take over driving, there is a possibility that they will collide unless an appropriate notification of transition is given. When drivers are engaged in something other than driving while automated driving system is controlling vehicle, it will be more probable that they don't understand the notification of transition given by the system.

In addition, some subjects could not understand what the alarm sound meant. It should be noted that among the subjects

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having 10 s to take over driving, there was one who became too much relieved because of enough distance from the road closer.

By reference to the above research achievements of JARI [13], we designed an HMI system to promote safe transition from automated to manual driving. We used not only an alarm sound and HUD but also voice guidance and interior lights as the tools of an HMI system in order to improve driver's understanding of the notification given by the system.

In addition, we expected that a step-by-step transition to manual would be effective for drivers to prepare for taking over driving well in advance. Therefore, we constructed two zones to prepare for taking over driving (preparation zone) on the route to a zone to take over driving (transition zone). In each zone, drivers were given notifications and instructions by the HMI system. The amount of time for drivers to take over driving in the transition zone was initially set by reference to the research results of JARI and was determined through a preliminary experiment.

The effect of the step-by-step transition supported by the HMI system with added tools was verified experimentally using a driving simulator (DS). With the DS, we constructed an experimental course which had three transition points. A questionnaire investigation and an interview were also conducted after the experiment to obtain subjective evaluation.

3.1 Situation to Request Drivers to Take over Driving

There is no clear definition or categorization of situations to request drivers to take over driving from automated driving. In this study, we assumed two situations. One is a situation which the system can detect well in advance by use of Intelligent Transport Systems (ITS). For example, an interchange (IC) on a highway corresponds to it. Another is a situation which the system detects with time pressing.

As the situation which the system can detect well in advance, we created an IC on a highway course. We also created a lane closer due to a traffic accident on a city road course with poor visibility. At this situation, the system cannot detect the accident scene until the vehicle turns the corner near the accident scene. Therefore, the system detects the accident scene with time pressing.

3.2 Experimental Equipment

We used a DS with the specification shown in Table 1. The subjects were seated in front of a display screen. They operated the simulator steering wheel, accelerator, and brake pedal as they watched the DS course displayed on the screen.

We also used an eye-gaze tracking system acquiring data of pupil areas which were recorded by means of a camera mounted in front of the subjects. The system records the data of pupil areas every 0.1 s. Another camera mounted near their

Table 1 Specification of the driving simulator

Function	Specification
Input device	Logicool GT Force Pro HPENVY
Vehicle motion Computing device	CPU: Intel Core i5 GPU: NVIDIA GeForce GTX 680
Image generation Device	Memory: 4GB
Output device	SHARP: AQUOS
Others	Seat: for SUBARU Legacy B4 (BE5D) Desk

feet recorded their foot motions. Table 2 shows the specifications of the eye-gaze tracking system.

3.3 Tools to Guide Drivers

As the tools to guide drivers, an alarm sound is the most widely used. However, it is not always easy for drivers to understand the meaning of the alarm correctly or instantly, as JARI reported that some drivers could not understand the meaning of the alarm sound [13]. Therefore, adding to the alarm sound and HUD, we used a voice guidance and interior lights as tools to guide drivers.

The alarm sound and interior lights have a role to call driver's attention. The voice guidance prompts drivers to do what they should do. The HUD displays the reason or the place that the transition to manual is required (Fig. 1). The HUD is projected at the lower right on the windshield. The interior lights are placed at a position lower than steering wheel so that drivers can look down and easily recognize it because young people tend to use smartphones very often (Fig. 2).

3.4 HMI System

We prepared two types of HMI systems. One is designed with four tools including a voice guidance, a alarm sound, HUD

Table 2 Specification of the eye-gaze tracking system

Function	Specification
Eye camera	SONY: CCD-TRV126 NTSC
Eye behavior interface unit	Techno Works: TE-9101B
Computing device	Dospara Prime CPU: Intel Core i5 Windows 7
Image generation device	GPU: Intel HD Graphics 4600 Memory: 2GB
Output device	MITSUBISHI: RDT 234WX



Fig. 1 Image of the HUD

and interior lights. It guides drivers in stages supplying preparation zones for transitions in addition to a zone to take over driving. Another is designed with alarm sound and HUD and gives driver only one signal to take over driving. The former is main object and is named HMI-M. The latter is comparative one to examine the effect of HMI-M and is named HMI-C.

These two HMI systems are classified into two categories depending on the situation. HMI-M is classified into HMI-M1 for the situation which the system can detect well in advance and HMI-M2 for the situation which the system detects with time pressing. HMI-C is also classified into HMI-C1 and HMI-C2.

The amount of time for drivers to take over driving in the transition zone was initially set by reference to the research of JARI [13]. The amount of time required to drive through other zones to prepare for transition was set initially by the authors. We drove on the experimental course repeatedly with the HMI system and figured out the initial amount of time.

After the initial amount of time was set, a preliminary experiment was conducted using two subjects who were young drivers aged 22 holding a driver's license. They were asked to drive on the course with the HMI system. We carefully observed their driving to evaluate whether the initial amount of time to drive through each zone was appropriate. We also obtained the evaluation of subjects on the timing of the notifications and instructions given by the HMI system. Judging from the objective evaluation and the subjective evaluation, the optimum amount of time to drive through each zone was determined with making fine adjustment if necessary.



Fig. 2 Interior warning light

3.4.1 HMI-M1 (Situation Detected Well in Advance)

HMI-M1 guides the driver to take back driving step-by-step through two preparation zones in addition to a zone to take over driving at the IC on highway (Fig. 3). One is to let the driver get back sense of driving. Another is to let him get back sense of speed.

The voice guidance is given as follows (Fig. 3).

- 1) The vehicle was controlled to run at 100 km/h by the automated driving system. When the vehicle comes to Point ① which is 30 s from the entrance of the IC (EI), HMI-M1 gives the first voice guidance to inform that the driver needs to take over driving after a short time at the IC. The voice guidance also prompts the driver to take a steering wheel and to monitor the surroundings. The specific content of the first voice guidance is “You will reach a transition point to manual driving before long. Please take the steering wheel and monitor the surroundings.”

The subjects received the explanation what the word “transition point” meant before the experiment.

- 2) The vehicle was controlled to run at 100 km/h by the automated driving system up to the deceleration lane. The second voice guidance is given at the starting point of the deceleration lane (Point ②) which is 12 s from the EI. The voice guidance informs the driver that the vehicle is being decelerated and that the point where he should take over driving comes soon. The vehicle is automatically decelerated from 100 km/h to 50 km/h in the zone from ② to ③. The content of the second voice guidance is “The vehicle is being decelerated. You will soon reach the transition point.”
- 3) At the point 7 s (at 50 km/h) from the entrance of the EI (Point ③), the third voice guidance prompts the driver to take over driving. The content of the third voice guidance is ‘You are at the transition point. Please take over driving.’
- 4) If the driver does not take over driving by the point 4 s (at 50 km/h) from the point where automated driving stops functioning (F), extra voice guidance is given to urge him to take over driving. The content of the extra voice guidance is “Please drive by yourself.” The volume of the voice guidance is larger than others.

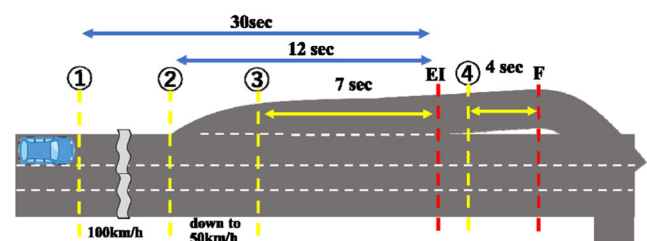


Fig. 3 Formation of HMI-M1

A zone from Point ① to Point ② is set to let the driver get back sense of driving. The drivers are expected to get back the sense of driving by monitoring surroundings with taking a steering wheel. Another zone from Point ② to Point ③ is set to let him get back sense of speed. The drivers are expected to get back the sense of speed by feeling that the vehicle is decelerating in addition to monitoring surroundings with taking a steering wheel.

The driver can take over driving by pushing a button. Even if he does not push the button, automated driving stops functioning at the prescribed finish point.

The alarm sound and the HUD are given at Point ①. The interior lights are also put on at Point ① and are put off when the driver takes over driving or when automated driving stops functioning mechanically.

As the alarm sound, sufficiently audible 1000 Hz buzzer sounds twelve times in 2.7 s. The HUD displays ‘highway IC’ where the transition to manual is required as shown in Fig. 1 in Section 3.3. The interior lights are red small lights lying side by side as shown in Fig. 2 in Section 3.3.

3.4.2 HMI-C1 (Situation Detected Well in Advance)

The vehicle was controlled to run at 100 km/h by the automated driving system until the deceleration lane. In the deceleration zone, the vehicle was automatically decelerated from 100 km/h to 50 km/h. HMI-C1 activates when the experimental vehicle comes to Point ① which is 7 s from the entrance of the IC (Fig. 4). It gives an alarm sound and HUD to prompt the driver to take over driving. The driver can take over whenever he wants. As the alarm sound, sufficiently audible 1000 Hz buzzer sounds twelve times in 2.7 s. The HUD displays ‘highway IC’ where the transition to manual is required.

3.4.3 HMI-M2 (Situation Detected with Time Pressing)

As the system detect this situation with time pressing, HMI-M2 supplied one preparation zone to let the driver recognize the situation for a transition to manual driving in addition to a zone for taking over driving. While being controlled by the automated driving system, the vehicle ran at 50 km/h.

A transition to manual driving must be finished by 20 m from the accident scene (B in Fig. 5). The distance from B to E was initially decided by reference to the research results of

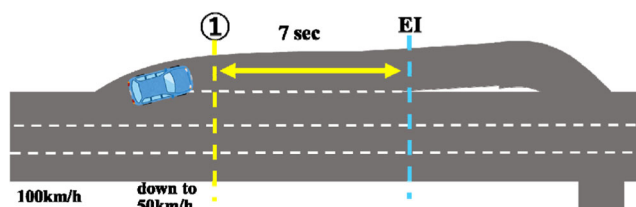


Fig. 4 Formation of HMI-C1

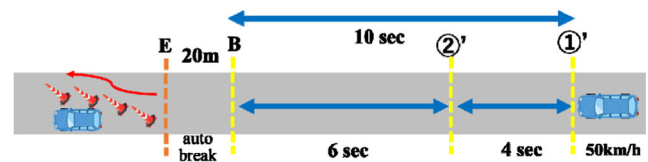


Fig. 5 Formation of HMI-M2

JARI [13]. And then, we determined through a preliminary experiment. If transition was not completed by the B, the vehicle is pulled up by an automatic brake system. The amount of time required to drive through from Point ②' to B was set 6 s (at 50 km/h). The amount of time required to drive through from Point ①' to Point ②' is 4 s (at 50 km/h).

The voice guidance is given as follows (Fig. 5).

- 1) When the vehicle comes to Point ①' which is 10 s (at 50 km/h) from B, HMI-M2 gives the first voice guidance to inform the driver about an obstacle ahead which automated driving cannot avoid to run through. The specific content of the first voice guidance is “There is an obstacle ahead.”
- 2) The second voice guidance is given at the point 6 s from B (Point ②') to prompt the driver to take over driving. The content of the second voice guidance is “Please take over driving.”
- 3) If the transition to manual driving is not completed by 20 m from the closer lane (B), the vehicle is stopped by an automatic brake system.

A zone from Point ①' to Point ②' is set to let the driver recognize the situation for a transition to manual driving.

The alarm sound and the HUD are given at Point ①'. As the alarm sound, sufficiently audible 1000 Hz buzzer sounds twelve times in 2.7 s. The HUD indicates that the obstacle is detected ahead. The interior lights are also put on at Point ①' and are put off when the driver take over driving or when the vehicle is stopped by the automatic brake system.

3.4.4 HMI-C2 (Situation Detected with Time Pressing)

While being controlled by the automated driving system, the vehicle ran at 50 km/h. HMI-C2 activates when the vehicle comes Point ①' which is 10 s (at 50 km/h) from B (Fig. 6). It gives the alarm sound and the HUD inducing the driver to take over driving. As the alarm sound, sufficiently audible 1000 Hz buzzer sounds twelve times in 2.7 s. The HUD indicates that the obstacle is detected ahead. As described earlier, the vehicle

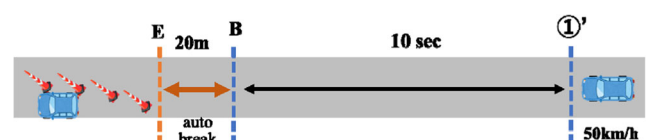


Fig. 6 Formation of HMI-C2

is stopped by the automatic brake system, if a transition to manual is not completed by 20 m (B) from the lane closer (E).

3.5 Experimental Course

We prepared an experimental course using a DS (Fig. 7). The experimental course includes a highway and a city road. The speed of the experimental vehicle is set at 100 km/h on the highway and 50 km/h on the city road, respectively. With these speeds, it takes 10 min to run through the highway and it takes 5 min to run through the city area, respectively.

A transition to manual from automated driving is required twice on the highway at Point 1 and Point 2 in Fig. 7. After getting off the highway, transition to manual from automated driving is required once at Point 3 on the city road. Before the start points where the driver is supposed to take over driving on the highway, the speed of the vehicle is automatically reduced to 50 km/h toward the entrance of the IC.

3.6 Driver's Task during Automated Driving Functions

Sonpo Holdings Inc. reported that conversation, eating and drinking are at the head of what drivers want to do in a vehicle controlled by an automated driving system [14]. We therefore asked the driver to talk to a fellow passenger (a researcher assumed a role) with eating and drinking while the vehicle was controlling by the automated driving system.

3.7 Questionnaire Investigation and Interview

We conducted a questionnaire investigation after the experiment to obtain the subjective evaluation data of the HMI systems. An overview of the investigation content is as follows. 1) Whether or not the subjects can take over driving smoothly, 2) Whether or not the timing of the voice guidance is appropriate, 3) Whether or not the tools to guide the subjects are appropriate.

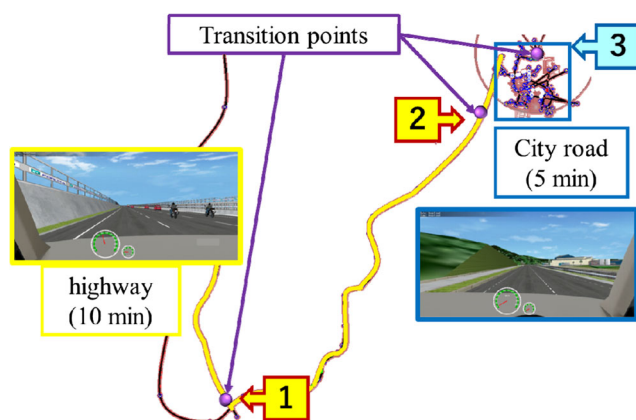


Fig. 7 Overview of the experimental course

We also made an interview to obtain further information and drivers' psychological state.

3.8 Subjects

Subjects were 41 young drivers from 18 to 23 years old holding a driver's license. Twenty-one beginner drivers were included. Among the rest of them, 4 drivers had over 2 year driving experience.

Automated driving technologies are expected to contribute to a reduction in accidents by controlling vehicles on behalf of mistake-prone human drivers. In order to decrease accidents drastically, drivers who are prone to cause accidents are expected to positively use automated driving technologies. Therefore, we believe that young drivers including many beginner drivers are present main targets as well as elderly drivers because their fatal accident rate is high. Thus, in this study, we prepared young drivers including beginner drivers as experimental subjects.

Twenty subjects drove with the assistance of HMI-M and other 21 drove with the assistance of HMI-C. The beginner drivers were divided into two groups almost equally. The drivers who had over 2 year driving experience were also divided into two groups equally.

In addition, according to the report of Nippon Research Center [15], young drivers in their twenties want to use an automated driving system more than other ages. On the other hand, they are concerned about the reliability of the technologies. In other words, young drivers need a support to use the automated driving system with confidence. Therefore, we chose them as our subjects to be supported with the HMI system.

3.9 Experimental Procedure

Before starting the experiment, we explained to the subjects on the significance of the study including that of the experiment. It was also explained that they were able to retire from participating in the experiment any time if they wanted. And then, they were asked to sign an acceptance form if they accepted the situation.

The subjects also received a detailed explanation about the function of the HMI systems. With the description of the formations of the HMI systems, we explained to them when each tool (alarm sound, HUD, voice guidance, interior lights) worked. We also explained to them on the meaning of the notification given by each tool. Regarding the voice guidance, we explained showing the contents of them on the paper.

The experiment was conducted according to the following steps. 1) Subjects drive on a test course, 2) The subjects drive on the experimental course with help of HMI-C or HMI-M, 3) A questionnaire investigation and an interview are made. On the test course, the subjects experienced automated driving

and practiced a transition to manual driving with the HMI systems.

Each subject's data was coded and anonymized. Data that could identify individuals was not stored.

3.10 Analysis of Data

A comparative analysis of the specific data between HMI-C and HMI-M was made to examine the effect of the HMI-M.

Data for analysis were pupil areas and the number of subjects who could not avoid a collision with the accident scene on the city road after they took over driving. As for the pupil areas, their mean values were compared. The increase-decrease rate of them was also analyzed. We also analyzed the time from when the subjects were prompted to take over driving until when they put their feet on the pedal.

On the highway course, the subjects were required to take over driving twice. At each time, the pupil areas were obtained at the following spots.

< HMI-C1>

- (a) Between 10 s and 5 s before the alarm sound was given.
- (b) From the timing when the alarm sound was given to the timing when the subjects took over driving.
- (c) From the timing when the subjects pushed the button to stop the function of automated driving and took over driving to the timing when the automated driving system activated again.
- (d) From the timing when the automated driving system activated again to 10 s after that.

< HMI-M1>

- (A) Between 10 s and 5 s before the first voice guidance was given.
- (B) From the timing when the first voice guidance was given to the timing when the second voice guidance was given.
- (C) From the timing when the second voice guidance was given to the timing when the third voice guidance was given.
- (D) From the timing when the third voice guidance was given to the timing when the subjects took over driving.
- (E) From the timing when the subjects took over driving to the timing when the automated driving system activated again.
- (F) From the timing when the automated driving system activated again to 10 s after that.

On the city road course, subjects needed to take over driving once. At the time, the specific data were obtained at the following spots.

< HMI-C2>

- (a) Between 10 s and 5 s before the alarm sound was given.
- (b) From the timing when the alarm sound was given to the timing when the subjects took over driving.
- (c) From the timing when the subjects took over driving to 10 s after that.

< HMI-M2>

- (A) Between 10 s and 5 s before the first voice guidance was given.
- (B) From the timing when the first voice guidance was given to the timing when the second voice guidance was given.
- (C) From the timing when the second voice guidance was given to the timing when the subjects took over driving.
- (D) From the timing when the subjects took over driving to 10 s after that.

For the analysis of the mean values of the time when the subjects put their feet on the pedal, we employed the Wilcoxon rank-sum test which is a nonparametric test handling data having no normal distribution. We used the chi-square test, which is also a nonparametric one, for the analysis of the number of subjects who made a collision with the accident scene. A text mining method was used on the responses of the interview to extract important phrases, and the context in which those phrases were embedded was used to create an organized summary of the important content.

4 Results and Discussions

4.1 Results of the Experiment

4.1.1 The Time from when the Subjects Were Prompted to Take over Driving until when they Put their Feet on the Pedal

HMI-C prompted the subjects to take over driving with the alarm and the HUD. HMI-M prompted them with the third voice guidance on the highway course and with the second voice on the city road course. We measured the specific time between I and II in Fig. 8. Figure 9 shows the foot position.

With the support of HMI-M, many subjects put their feet on the pedal when they were given the first voice guidance which prompted the subjects to take the steering wheel and to monitor the surroundings for the transition to manual driving after a short time. Therefore, their feet were on the pedal already when they were prompted to take over driving. Consequently, the measurement values for those subjects were zero.

Analysis showed that the mean values of the time when the subjects put their feet on the pedal were significantly shorter

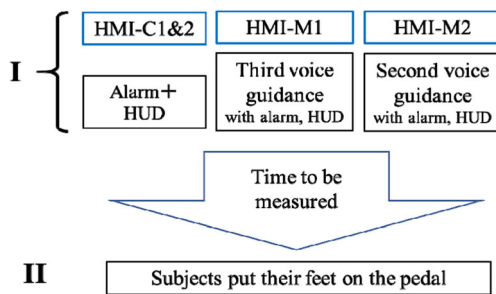


Fig. 8 Time to be measured

with the support of the HMI-M than with that of the HMI-C, at all spots (Fig. 10).

It is indicated that a stepwise transition to manual driving with the support of the HMI-M is effective for drivers to take over driving more smoothly. It is also indicated that specific instructions by the voice guidance promotes drivers' understanding of the situation.

4.1.2 Pupil Areas

Figure 11 shows the increase-decrease of the pupil areas at the first transition to manual driving on the highway. They show noteworthy feature.

A to F on the horizontal axis of Fig. 11 mean the zones to obtain the pupil area, as explained as follows.

- From 10 to 5 s before the HMI systems start to activate.
- From the timing when the first voice guidance of the HMI-M1 inform that the driver needs to take over driving after a short time until the timing when the second one informs vehicle's deceleration.
- From the timing when the second voice guidance of HMI-M1 is given until the timing when the third one prompts the subjects to take over driving.
- From the timing when the voice guidance of both HMI-M1 and HMI-C1 prompts the subjects to take over driving until the timing when the subjects took over driving.
- From the timing when the subjects accomplish transition until the timing when the automated driving system activate again.
- From the timing when the automated driving system activate again to 10 s after that.

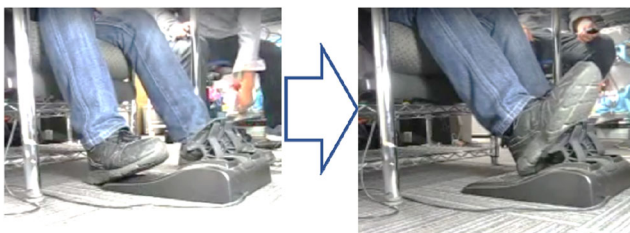


Fig. 9 Picture of foot position

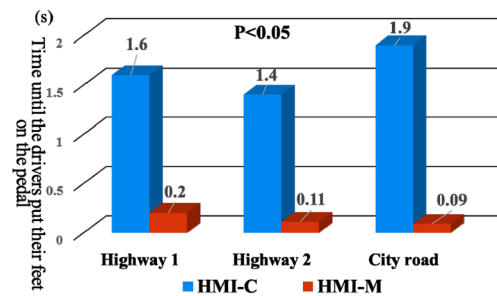


Fig. 10 Average of time from the timing when the HMI systems prompt to take over driving to the timing when they put their feet on the pedal

As for the pupil areas of the subjects who were supported by HMI-M1, they gradually increased and peaked at zone D where the driver was expected to take over driving. On the other hand, those of the subjects who were supported by HMI-C1, suddenly increased but peaked after transition was accomplished.

When a person heightens concentration, his pupil tends to expand. Thus, the results indicate that the subjects with HMI-M1 gradually increased their concentration and took over driving in the most concentrated state, though those with HMI-C1 took over driving before their concentration became highest. It is suggested that the step-by-step guidance by HMI-M1 is more effective for drivers to take over driving readily with sufficient concentration.

The similar tendency was found at the second transition on the highway.

Figure 12 shows the increase-decrease of the pupil areas at the transition on the city road.

A to D on the horizontal axis of Fig. 12 mean the zones to obtain the pupil area, as explained as follows.

- From 10 to 5 s before the HMI systems start to activate.
- From the timing when the first voice guidance of HMI-M2 informs that the driver needs to take over driving after a short time until the timing when the second voice guidance prompts the subjects to take over driving.
- From the timing when the voice guidance of both HMI-M2 and HMI-C2 prompts the subjects to take over

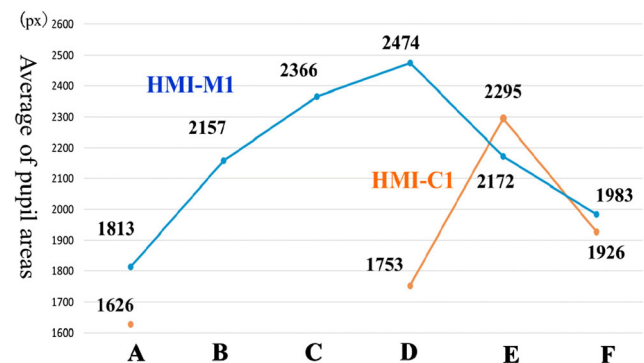


Fig. 11 Increase-decrease of the pupil areas at the first transition to manual driving on the highway

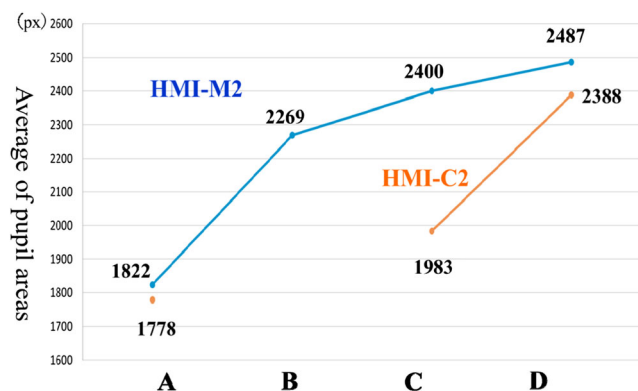


Fig. 12 Increase-decrease of the pupil areas at the transition to manual driving on the city road

driving until the timing when the subjects accomplish transition.

- D) From the timing when the subjects accomplish transition to 10 s after that.

At the transition on the city road, both the pupil areas of the subjects supported by HMI-M2 and those of them supported by HMI-C2 peaked after transition was accomplished. The results indicate that one preparation zone supplied by HMI-M2 was not enough to make a readily transition. It is suggested that, for more smooth and safer transition, drivers need the step-by-step guidance as HMI-M1 gives which is enabled with ITS.

4.1.3 Number of Collisions

We counted the number of subjects who made the vehicle collision with the accident scene after they took over driving on the city road. This number includes the case where the vehicle entered the accident area and the case where the vehicle hit the wall on the opposite side of the road because the subject turned the steering wheel too much to dodge the accident scene.

As Fig. 13 shows, 19 out of 21 subjects supported by HMI-C2 made a vehicle collision after they took over driving. This

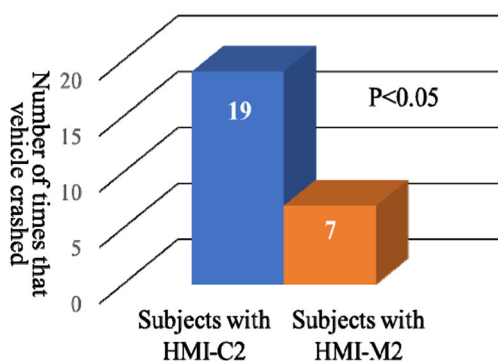


Fig. 13 Number of subjects who made collisions at the transition on the city road

number is significantly higher compared to the case with the subjects supported by HMI-M2. It can be said that transition supported by HMI-M2 is safer than that supported by HMI-C2.

There was no case that the vehicle was stopped by the automatic brake system.

4.2 Results of the Questionnaire Investigation and Interview

4.2.1 Questionnaire Investigation

Figure 14 shows the ratio of respondents to the question asking whether the subjects took over driving smoothly with the supported by HMI-M1 or HMI-C1.

Explanatory notes ① to ④ in Fig. 14 mean answers to be chosen as explained as follows.

Transition to manual is

- ① carried out smoothly
- ② carried out somewhat smoothly
- ③ not carried out very smoothly
- ④ not carried out smoothly at all

Eighty percent of the subjects supported by HMI-M1 feel their transition smooth. In contrast, less than half of the subjects supported by HMI-C1 feel so. Further, some subjects with HMI-C1 do not feel quite smooth. These results indicate that HMI-M1 is more effective to make a smooth transition than HMI-C1.

Evaluation of the second transition on the highway has a similar tendency (Fig. 15).

Further, in terms of the subjects who feel that their transition was made smoothly, the proportion of the subjects with HMI-M2 on the city road (Fig. 16) is smaller than that of those with HMI-M1 on the highway (Figs. 14 and 15), especially that of subjects who chose answer ① is smaller.

These results are corroborative with those of the experiment indicating that the step-by-step guidance given by

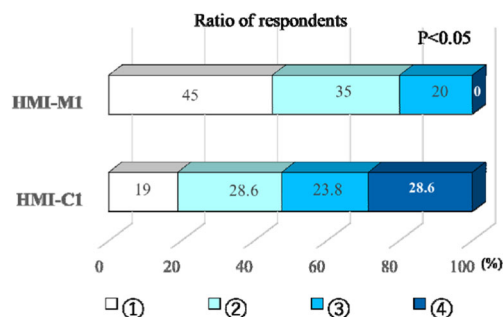


Fig. 14 Evaluation of the effect of HMI-M1 and HMI-C1 to make the first transition smoothly on the highway

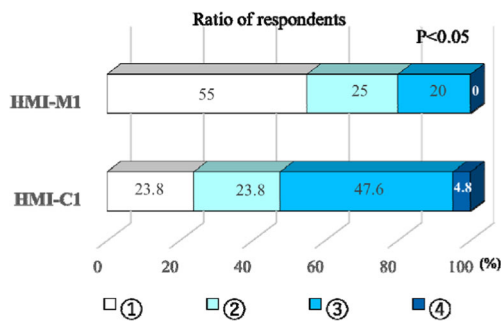


Fig. 15 Evaluation of the effect of HMI-M1 and HMI-C1 to make the second transition smoothly on the highway

HMI-M1 is more effective for drivers to take over driving readily with concentration.

The second question is about the timing of the guidance given by the HMI systems.

Figure 17 shows the evaluation of the timing of the guidance provided by HMI-M1 and HMI-C1 at the first transition on the highway. Data were analyzed by chi-square test. Explanatory notes ① to ⑤ in Fig. 17 mean answers to be chosen as explained as follows.

Timing is

- ① too early
- ② a little early
- ③ good
- ④ a little late
- ⑤ too late

Evaluation of the timing of HMI-M1 is significantly high. There is no subject who chose the answers ①, ④ and ⑤. On the other hands, about 40% of subjects feel that the timing of HMI-C1 is late.

As Fig. 18 shows, evaluation of the timing at the second transition on the city road has a similar tendency, although no significant difference was found (chi-square test).

Comparing the evaluation of the timing on the highway (Fig. 17) and that on the city road (Fig. 18), the former is

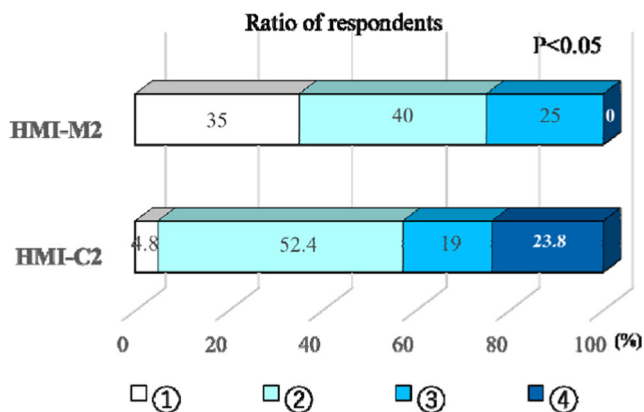


Fig. 16 Evaluation of the effect of HMI-M2 and HMI-C2 to make the second transition smoothly on the city road

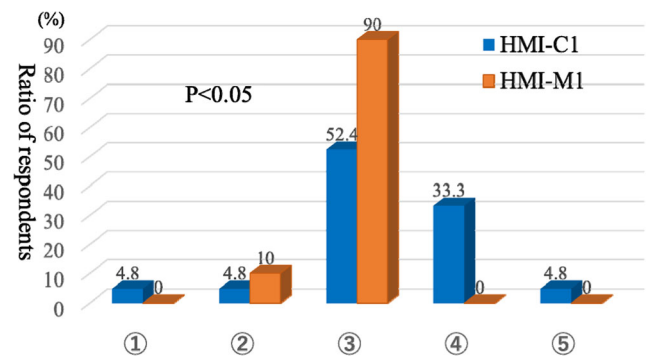


Fig. 17 Evaluation of the timing of the guidance given by HMI-M1 and HMI-C1 at the first transition on the highway

higher than the latter for both type of the HMI systems. It is indicated that the guidance provided at the situation which the system detected well in advance the place where transition to manual was required, was easier for drivers to follow than that provided at the situation which the system detected with time pressing.

The third question is to ask whether the tools to guide the subjects are appropriate. There are four tools, voice guidance, alarm sound, HUD and interior lights. In order to evaluate each tool, subjects were asked to choose one answer from the five options as follows.

- ① helpful
- ② somewhat helpful
- ③ no matter whether it is present or absent
- ④ rather helpless
- ⑤ helpless

As shown in Fig. 19, the voice guidance is highly appreciated. All subjects admit the preferable effect of the voice guidance.

We also asked the subjects who drove with HMI-C1 & C2 if they wanted the voice guidance. Almost 90% of them answered that the voice guidance was desirable support.

The HUD also obtained a good evaluation from both the subjects with HMI-M1 & M2 and those with HMI-C1 & C2

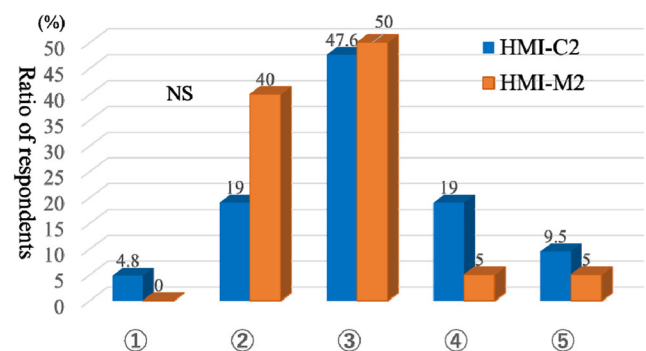


Fig. 18 Evaluation of the timing of the guidance given by HMI-M2 and HMI-C2 at the transition on the city road

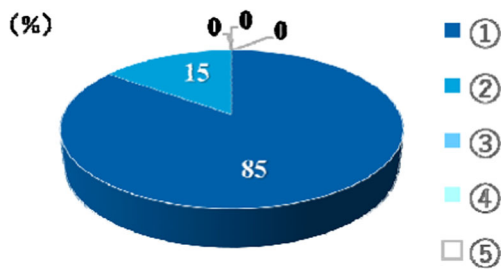


Fig. 19 Evaluation of the voice guidance given by HMI-M1 & M2

(Fig. 20). We found no subject who felt that the HUD was an obstacle for their vision, so the HUD can be admitted appropriate as a means for checking the situation.

Figure 21 shows the evaluation of the alarm sound. Both the subjects with HMI-M1 & M2 and those with HMI-C1 & C2 admitted its helpfulness.

The interior lights attached to the vehicle with HMI-M1 & M2 also obtained a good evaluation (Fig. 22). Some subjects were raised their attention strongly by the interior lights.

It is indicated that the alarm sound and the interior lights are appropriate to raise attention of drivers.

4.2.2 Interview

By the analysis of the responses to the interview, some important phrases were extracted. With the important phrases and the context in which those phrases were embedded, we summarized as follows.

From the responses of the subjects with HMI-C1 & C2, it is shown that many subjects took over driving without getting back the sense of driving including the sense of speed, steering and controlling the pedals. This indicates that the subjects could not take over driving readily. It corroborates the results of the experiment.

In terms of steering and controlling the pedals, some subjects with HMI-M1 & M2 also admitted that they drove without getting back those sense. Although HMI-M1 & M2 supply zones for drivers to get back the sense of driving and of speed, it is presumed that those two zones are insufficient for drivers to regain their sense of steering and of controlling the pedals. It is speculated that the major cause of the collisions of the vehicle on the city road might have been that the subjects could not regain those feelings.

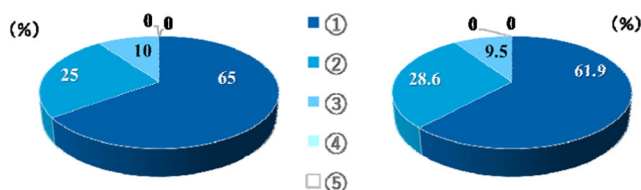


Fig. 20 Left graph shows the evaluation of the subjects with HMI-M1&M2 on the HUD and right one shows the evaluation of those with HMI-C1&C2 on it

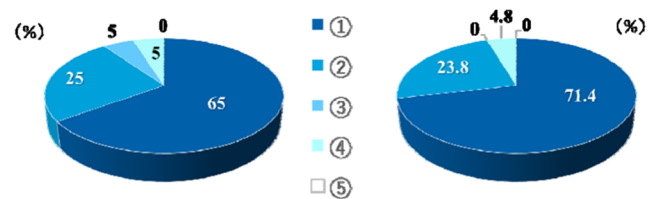


Fig. 21 Left graph shows the evaluation of the subjects with HMI-M1&M2 on the alarm sound and right one shows the evaluation of those with HMI-C1&C2 on it

The responses of the subjects with HMI-C1 & C2 show that the subjects want the first notice regarding to transition earlier. In addition, from the responses of the subjects with HMI-M2, it is shown that the time to make the transition on the city road (from the timing when the first voice guidance is given until the timing when the second voice guidance prompts transition) is too short to prepare smooth transition. These results confirm that the step-by-step guidance as HMI-M1 supplies is necessary for drivers to make smooth and safe transition.

In terms of useful tools, the followings are shown.

Both the voice guidance and the HUD are effective, receiving a high evaluation in the investigation. Further, information given by the combination of the voice guidance and the HUD was admitted much easier to understand.

It is also confirmed that the interior lights placed at a position lower than the steering wheel do not disturb driving.

Some subjects complained that they were a little startled and nervous by the alarm sound. There may be important to study to identify alarm sounds which do not startle any driver.

5 Conclusion

We designed an HMI system (HMI-M) promoting smooth and safe transition to manual driving from Level 3 automated driving. The HMI system induces drivers with using a voice guidance, an alarm sound, HUD and interior lights. On the route to the transition zone, HMI-M1 supplies drivers with two preparation zones, one to regain driving sense and the other to regain sense of speed. HMI-M2 supplies a preparation zone to recognize the situation properly.

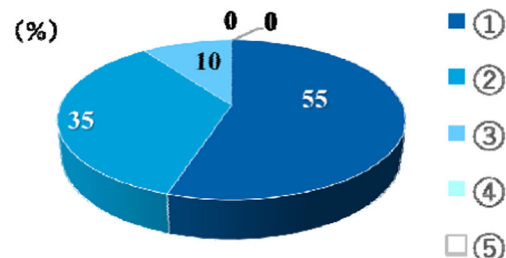


Fig. 22 Evaluation of the interior lights given by HMI-M1&M2

Through an experiment and a questionnaire investigation, it is indicated that HMI-M is more effective to promote smooth and safe transition than HMI-C using only alarm and HUD. Particularly, HMI-M1 was clarified to be very effective. It is suggested that voice guidance is an effective tool for drivers to prepare transition and that interior lights are helpful for drivers to notice the request of transition. It is also suggested that a step-by-step transition with two preparation zones is important for drivers to take over driving readily with concentration. Further, it is considered that fully equipped ITS is important to enable detection of the necessity of transition to manual driving well in advance and to provide enough time for drivers to prepare a transition.

6 Future Study

In the present study, two specific zones were supplied for drivers to regain driving sense and sense of speed. We verified that these zones had a certain effect. However, it is also suggested that they will not fulfill a role to let drivers regain the sense of steering and of controlling the pedals. Therefore, it is necessary to study an HMI system to let drivers regain the sense of steering and of controlling the pedals [16].

Further, it is crucial to study countermeasures in case that the driver does not notice the signal requesting transition to manual driving because he is sleeping or using a smartphone [17] with earphones or any other reasons.

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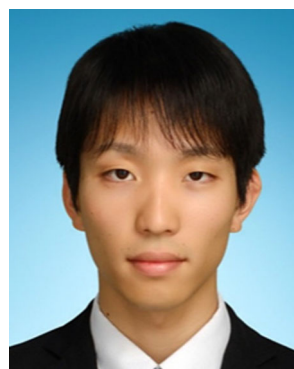
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